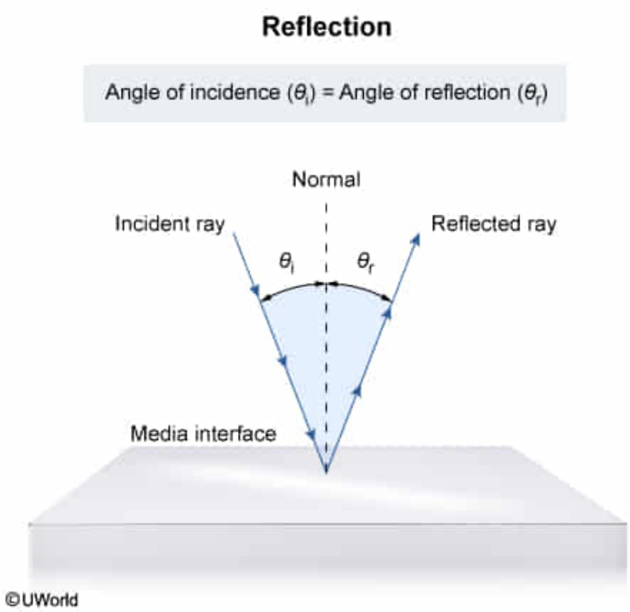
Index of refraction, old medium (n_1)	Angle of incidence (θ_i)
1.2	24°
1.6	18°
2.4	12°

The table above gives the angle of incidence θ_i for light encountering a new medium at three unique smooth interfaces. The refractive index of the new medium is identical in each case. Some light reflects off the media interface in each condition. How does the angle of reflection θ_r in the high refractive index condition (large n_1) compare to the angle of reflection in the low refractive index condition (small n_1)?

- A. θ_r will be 1/4 as great. [5%]
- B. θ_r will be 1/2 as great. [55%]
- C. θ_r will be approximately equal. [20%]
- D. θ_r will be 2 times as great. [19%]

Correct
55% answered correctly

Explanation:



Reflection is a phenomenon in which light that arrives at the interface between two media re-enters the original media with modified direction. Models of reflection assume that light is comprised of individual rays rather than dispersed wavefronts. This simplification allows for accurate predictions about the pathway that light will take during reflection events.

Specifically, the pathway of light following a reflection event is described in terms of the angle that light rays make with the normal, an imaginary line perpendicular to the interface between media. For example, the **angle of incidence (θ_1 or θ_i)** refers to the angle (relative to the normal) at which light approaches the interface between two media.

When light strikes a media interface, some light may pass through the first medium, enter the second medium, and undergo refraction as governed by Snell's law. However, the remaining light does not enter the second medium and instead reflects back into the first medium at the **angle of reflection (θ_r)**.

One principle characteristic of reflection is that the angle of reflection is always equal to the angle of incidence, irrespective of the refractive index on either side of the media interface:

$$\theta_i = \theta_r$$

Consequently, light rays approaching equally smooth interfaces at different angles of incidence will reflect away from each interface at different angles of reflection. Therefore, light rays that approach a smooth interface at an angle of 12° (high refractive index condition) will reflect away from the interface at 12°, which is half the angle of light rays approaching the interface at an angle of 24° (low refractive index condition) **(Choice B)**.

Educational objective:
Light reflects from an interface between media and travels in a new direction within the original medium. For smooth media interfaces, the angle of reflection is equal to the angle of incidence.

Time spent:0 sec
Subject:
Physics

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System:
Light and Sound